

CSCI 7350 Data mining

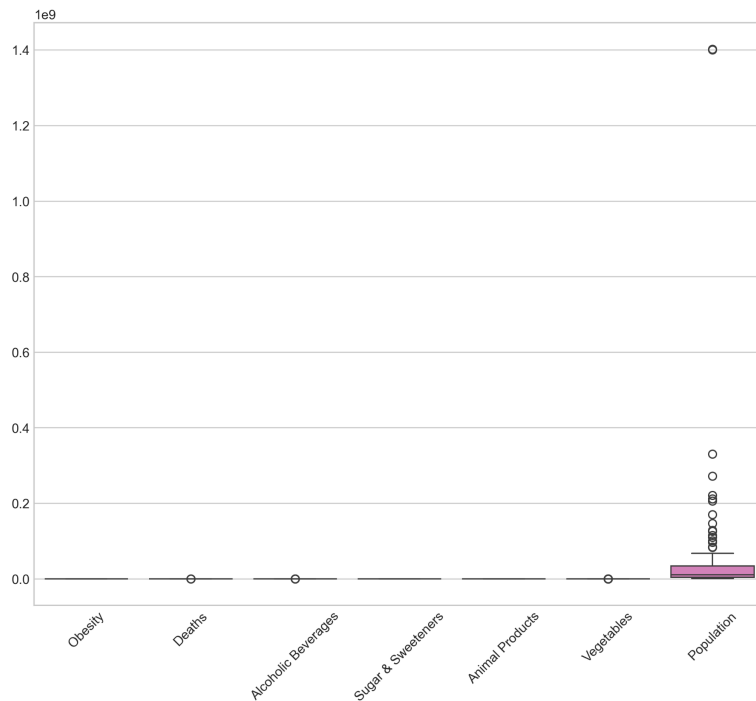
Comprehensive exam report

Wasim Raja Mondal

Dataset

We have chosen Covid-19 Healthy Diet Dataset (original) dataset for this work.

Part 1: Data Understanding & Preprocessing



From the figure, it is immediately clear that **population exhibits a very large scale and extreme outliers**, with a few countries having exceptionally high population values compared to the rest. This causes the population feature to dominate the scale of the plot, making the variation in other variables appear compressed near zero. The presence of many outliers in population indicates a **highly skewed distribution**, which will likely require transformation (such as logarithmic scaling) before modeling.

Part 2: Feature Selection & Engineering

We have chosen obesity as the target variable. Feature selection was performed using the **Variance Threshold** method, **two-sample t-test**, **chi-square ((chi2)) test**, **SelectKBest**. The selected features are shown using these methods in the notebook.

Part 3: Supervised Learning

Logistic Regression performance comparison

Feature Set	Accuracy	Precision	Recall	ROC-AUC	PR-AUC
Original	0.809	0.905	0.731	0.912	0.861
Variance Threshold	0.809	0.905	0.731	0.912	0.861
Mutual Information	0.936	0.960	0.923	0.954	0.969
T-test	0.915	0.923	0.923	0.952	0.970
Chi-square	0.809	0.815	0.846	0.925	0.939

Comparative Performance Analysis of Decision Tree Classifier Using Various Feature Selection Methods

Feature Set	Accuracy	Precision	Recall	ROC-AUC	PR-AUC
Original	0.872	0.955	0.808	0.878	0.874
Variance Threshold	0.745	0.792	0.731	0.737	0.715
Mutual Information	0.936	0.960	0.923	0.958	0.972
T-test	0.915	0.923	0.923	0.952	0.970
Chi-square	0.809	0.815	0.846	0.925	0.939

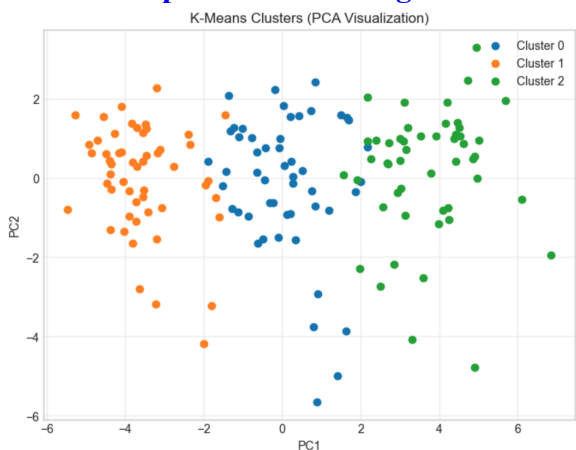
Comparative Performance Analysis of Random Forest Classifier Using Various Feature Selection Methods

Feature Set	Accuracy	Precision	Recall	ROC-AUC	PR-AUC
Original	0.872	0.917	0.846	0.967	0.976
Variance Threshold	0.851	0.880	0.846	0.954	0.967
Mutual Information	0.872	0.857	0.923	0.965	0.974
T-test	0.872	0.885	0.885	0.956	0.966
Chi-square	0.830	0.821	0.885	0.956	0.970

Comparative Performance Analysis of Support Vector Machine (SVM) Classifier Using Various Feature Selection Methods

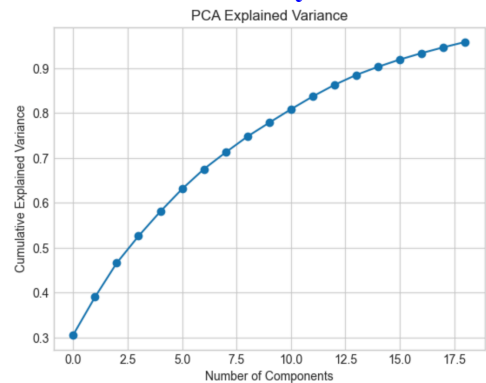
Feature Set	Accuracy	Precision	Recall	ROC-AUC	PR-AUC
Original	0.872	0.917	0.846	0.934	0.946
Variance Threshold	0.809	0.840	0.808	0.929	0.949
Mutual Information	0.851	0.828	0.923	0.945	0.956
Chi-square	0.830	0.846	0.846	0.947	0.963

Part 4: Unsupervised Learning



The PCA visualization of K-Means clustering reveals three distinct groups of countries, primarily separated along the first principal component, indicating that the dominant variation in dietary and health-related features drives cluster formation. One cluster consists of countries with lower dietary intensity (negative PC1), another represents moderate or mixed dietary patterns, and the third cluster corresponds to countries with higher consumption of calorie-dense and animal-based products. Some overlap between clusters suggests gradual transitions rather than strict separations, while a few outliers indicate unique dietary behaviors.

Part 5: Dimensionality Reduction



Principal Component Analysis (PCA) was applied after feature standardization to reduce dimensionality while preserving variance. The cumulative explained variance plot shows a gradual increase, indicating that information is distributed across multiple features. Approximately 19 principal components were required to retain 95% of the total variance, reducing the feature space from 29 to 19 dimensions. This suggests moderate feature redundancy, but also highlights the intrinsic complexity of the dataset, where multiple variables contribute to the overall variance.

Part 6: Experimental Design & Analysis

